



# Augmentor: a cognitive-inspired approach for context-aware adaptive learning with human guidance

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## Abstract

Automation in complex industrial processes faces significant challenges due to its heavy reliance on human knowledge and tacit expertise. This knowledge is typically acquired through years of on-site training, making the transition to automation difficult. While data-driven approaches offer potential solutions by building automation models from expert observations, they often fall short of capturing deeper human understanding and contextual reasoning. In this paper, we introduce Augmentor, a cognitive-inspired framework for context-aware adaptive learning. The framework bridges reactive learning, which continuously updates models based on incoming data, with context-aware adaptive learning, which incorporates external context and expert knowledge for higher-order reasoning and decision-making. Inspired by theories of consciousness, Augmentor leverages autonomic computing principles through the MAPE-K model and semantic web technologies to create a global workspace for managing knowledge and guiding adaptive learning processes. The approach addresses two key objectives: continuously refining models through reactive learning and enabling experts to inject contextual knowledge to guide the learning process. Human-Gated Data Aggregation is used for data-driven updates, while context-aware adaptive learning ensures the system dynamically adjusts to complex scenarios and evolving industrial conditions. We validate Augmentor through a real-world use case in collaboration with our industrial partner, Mekanisk Service Halden, which specialises in various mechanical services. Experimental results show significant performance improvements across three strategies—static model without relearning, reactive learning, and context-aware adaptive learning—demonstrating the advantages of incorporating context-aware strategies to achieve more robust and adaptive control.

**Keywords** Smart manufacturing · Artificial intelligence · Learning system

## Introduction

The industrial sector has undergone four major revolutions, each driven by the pursuit of greater productivity through automation. As automation replaced many manual and

repetitive tasks, human roles shifted toward more complex, cognitive responsibilities. However, certain tasks that require nuanced decision-making and adaptability remain beyond automation's reach. In Computer Numerical Control (CNC) machining, automation has advanced significantly, yet human operators are still essential for machine setup, monitoring, and corrective actions. While Industry 4.0 technologies focus heavily on optimizing workflows and processes such as condition monitoring, anomaly detection, and process control (Zheng et al., 2021), Mohanraj et al. (2021), they often overlook the value of human expertise.

The concept of learning from humans and transferring that knowledge to machines is not new; it predates Industry 4.0 and has long been a core concern in artificial intelligence. Learning from humans can be approached in two main ways: the intelligence design approach and the learning from examples (Kambhampati, 2021). The intelligence design approach involves explicitly modeling human knowledge

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and encoding it into formal representations such as documentation, ontologies, or Bayesian networks (Rosário et al., 2015). In manufacturing, this method has been more commonly applied to high-level organizational or strategic knowledge, rather than to the operational insights found on the shop floor (Fenoglio et al., 2022). At the shop floor level, learning from examples is more popular. It involves capturing human behavior patterns through data-driven techniques like imitation learning (Nguyen et al., 2023), Zhu and Hu (2018), enabling machines to replicate operator actions. However, these methods primarily extract behavioral patterns and often miss the underlying reasoning, context-awareness, and nuanced decision-making that skilled operators apply in real-world settings. Both approaches fall short of fully learning how experienced operators think and act. Skilled human operators integrate formal knowledge with situational awareness and hands-on experience. They analyze complex, dynamic situations, connect their actions to broader organizational goals, and apply refined control strategies. This deeper, more contextual understanding to drive learning is still difficult to replicate in current machine learning paradigms, underscoring the need for systems to better incorporate human expertise as a central component rather than a peripheral consideration.

To address this, we propose the Augmentor system, a human-centric approach that fosters knowledge exchange and collaboration between humans and machine learning algorithms. Our system is designed to emulate human training processes by achieving the following objectives:

1. *Objective 1* To continuously refine the learning algorithms by observing and letting operators adjust their decisions.
2. *Objective 2* To incorporate contextual information, environmental understanding, and operator knowledge into the learning system.

Drawing inspiration from cognitive science and theories of machine consciousness, Augmentor introduces a framework for context-aware adaptive learning, defined as continuous learning enhanced by contextual information. Unlike traditional machine learning approaches, which are typically driven by data-centric, context-aware adaptive learning incorporates external knowledge and contextual reasoning to guide the learning process. The framework brings together advances in artificial intelligence and autonomic computing, while also integrating key elements from cognitive science, to achieve the two core objectives above by:

1. Facilitating context-aware adaptive learning from data through human-guided mechanisms.

2. Utilizing the Monitoring - Analysing - Planning - Execution - Knowledge (MAPE-K) framework as the foundation for a global workspace, aligned with theoretical perspectives on machine consciousness.

We present a practical use case in CNC manufacturing developed in collaboration with Mekanisk Service Halden (MSH),<sup>1</sup> a company specializing in mechanical services such as welding, industrial maintenance, and parts machining. The case addresses the growing challenge of labor shortages in the CNC sector. Scaling up production requires skilled operators, but traditional training often takes 5 to 7 years due to the tacit nature of machining expertise, which is acquired primarily through hands-on experience and is difficult to formalize. We collected sensor data from a real industrial CNC machine, Mazak Integrex i500 series. Augmentor learns from experienced operators by observing their control strategies, such as adjustments to the linear feed rate override. By integrating adaptive policies into its knowledge base in real time, the system accelerates its learning process and improves its responsiveness. Our findings demonstrate that expert human guidance significantly enhances learning efficiency, reduces error rates, and improves system performance. By capturing control patterns directly from operators, Augmentor aims to modernize traditional monitoring processes and support operators in overseeing multiple machines or focusing on higher-level tasks. Through skill-level-adapted decision support, the system contributes to increased productivity, reduced production errors, and the development of a more resilient and capable workforce.

The structure of this paper is as follows: Sect. 2 introduces our industrial use case in CNC machining, which serves as both the inspiration and validation for our work. Section 3 outlines key foundational concepts. Section 4 details the Augmentor framework, grounded in the principles of machine consciousness. Section 5 demonstrates the application of Augmentor within our industrial use case. Section 6 presents the results of our experiment, while Sect. 7 analyses these findings and explores actionable insights within a broader application context. Section 8 concludes the paper with a summary of findings and potential future directions.

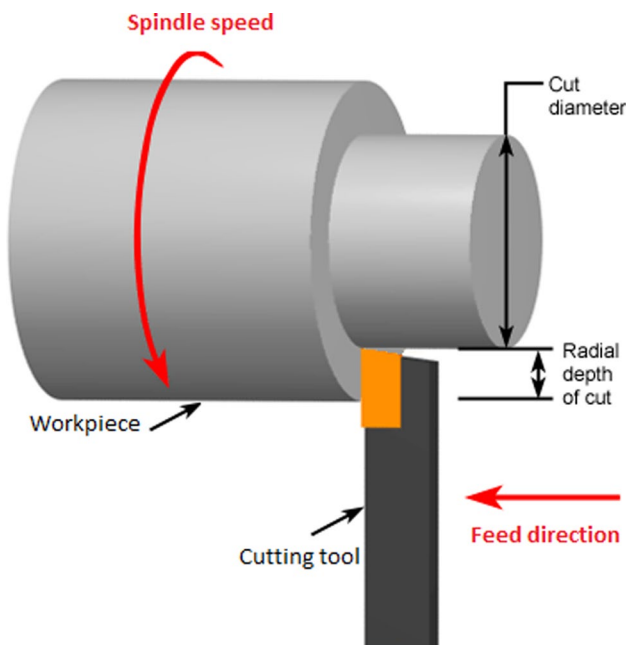
## Motivation scenario

As part of its production expansion, MSH has acquired several state-of-the-art CNC machines, including the Mazak Integrex i500 series. These machines are equipped with modern sensor technology that supports the monitoring

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**Fig. 1** The control interface for Mazak Integrex i500. The yellow box shows the current commands. The green box displays cutter positions. The pink box shows cutter loads. The blue box contains knobs to adjust engagement rates



**Fig. 2** Turning spindle speed and feed rate. Figure adapted by the authors from CustomPartNet: speed-feed-turning (2008)

process. Figure 1 shows the control interface displaying sensor data from the CNC machine, allowing operators to monitor the machine's status and make informed decisions to adjust and optimize the process.

Our research leverages the digitalization of CNC machine data through sensors and operator interactions with the control interface to develop a learning algorithm capable of predicting when operator intervention is necessary. One key action we focus on is the adjustment of the machine's linear feed rate, which controls how fast the tool moves in the linear directions ( $x$ ,  $y$ ,  $z$ ) within the Cartesian coordinate system, which is shown in Fig. 2. In CNC machines, the feed rate is predefined in the CNC code. For example, the code `G1 Z-1.017 F1500` directs the tool to move along the  $Z$ -axis at a feed rate of 1500 units per minute. However, real-world conditions often require custom adjustment. For example, if the raw material surface is too rough, using the predefined feed rate might be unsafe, necessitating a customized speed adjustment for each specific workpiece. Similarly, tool wear conditions may also demand feed rate changes. Feed rate override ( $Fovr$ ) allows operators to adjust the feed rate using knobs and buttons on the control interface (Fig. 1). At 100%, the machine operates at the speed set by the CNC program, while reducing  $Fovr$  to 0% halts the machine. This ability to manually adjust  $Fovr$  underscores the importance of operator expertise, as human judgment and flexibility are crucial for making nuanced decisions in response to varying production conditions.

The Mazak Integrex i500 is equipped with several built-in sensors that stream data via the MTConnect protocol in XML format. While there are a total of 127 signals, our hypothesis assumes that only the signals available on the control interface, which operators use for monitoring, serve as key information sources influencing their decisions. Therefore, we selected 20 relevant signals, as shown in Table 1, to use as input for our models. These signals can be categorized into four main groups, each reflecting critical properties of the cutting tools during the machining process.

1. Linear and rotational positions of the cutting tool and workpiece, which provide information about the geometry and dimensions of the machined part.
2. Linear and rotational feed rates, which describe the machine's motion.
3. Spindle and axis load data, which can offer insights into the material properties, surface finish quality, and the condition or wear level of the cutting tool.
4. Temperature is regulated to remain stable during the machining process. However, different materials have different optimal temperature ranges for effective and safe machining.

**Table 1** List of input sensor signals

| Signal  | Unit    | Description                           |
|---------|---------|---------------------------------------|
| X       | mm      | Linear positions of the cutting tools |
| Y       | mm      |                                       |
| Z       | mm      |                                       |
| B       | degree  | B angle of the cutting tools          |
| C       | degree  | C angle of the workpiece              |
| X_frt   | mm/s    | Linear feed rate of the cutting tools |
| Y_frt   | mm/s    |                                       |
| Z_frt   | mm/s    |                                       |
| MS_rpm  | rpm     | Milling spindle's rotary velocity     |
| TS_rpm  | rpm     | Turning spindle's rotary velocity     |
| X_load  | %       | Linear load on the cutting tools      |
| Y_load  | %       |                                       |
| Z_load  | %       |                                       |
| MS_load | %       | Milling spindle's load                |
| TS_load | %       | Turning spindle's load                |
| X temp  | celsius | Temperature of X servo motor          |
| Y temp  | celsius | Temperature of Y servo motor          |
| Z temp  | celsius | Temperature of Z servo motor          |
| MS temp | celsius | Temperature of milling spindle motor  |
| TS temp | celsius | Temperature of turning spindle motor  |

We define the objective function for behavioral cloning in this study as follows:

$$y_{t+g} = \pi(X_t)$$

in which:

- $X_t \in \mathbb{R}^{N \times M}$  is a vector  $X_t = (x_t, x_{t-1}, \dots, x_{t-N})^T$ ,  $x_t$  is a vector with M features at timestep  $t$ ; In this case,  $M = 20$  which are the 20 feature signals shown in Table 1. The feature signals contain information regarding movements, temperatures, and loads on the cutting tools. They match the information on the control interface (Fig. 1). N is the window size.
- $y_t$  is a vector  $y_t = Fovr_t$ ;  $g$  is the time into the future that the model forecasts. The AI models predict operators' decisions after the  $g$  time gap. This is to prevent shortcut learning and to accommodate the decision support system.  $g$  is set to one second in the experiment.

In our previous work, we focused on investigating imitation learning (Hussein et al., 2017) as a way to learn how to control *Fovr* from experienced operators (Nguyen et al., 2023), Nguyen et al. (2024). In these works, we used behavioral cloning (Bratko et al., 1995), the simplest and most straightforward form of imitation learning, to quickly capture the core patterns in the dataset we collected from the Mazak machine. Our previous work focused on exploring batch learning using data collected from one month of production at MSH. During this time, MSH produced several instances of the same products. This reduced the amount

of variabilities and allowed us to examine the feasibility of using imitation learning to learn the control patterns from data collected from operators. Our results showed that behavioral cloning is capable of capturing the control patterns from the collected dataset. Even though ensemble models such as Random Forest were comparative to Long Short Term Memory (LSTM), we opted to use the LSTM model acquired from the first phase in this study. This is because LSTMs are built to handle temporal dependencies and are more suitable for sequential modeling, which is our use case.

Behavioral cloning carries a significant risk of distributional shift, which occurs when the collected dataset fails to represent the full range of possible scenarios (Hussein et al., 2017). In our previous study, we developed a seed model that captured certain control patterns within a limited set of environmental variations. To mitigate distributional shifts, it is essential to collect more data across a broader range of scenarios to better account for variability. In manufacturing, with CNC machining as an example, both knowledge and process variability are constantly evolving, driven by new products and changing requirements. As a result, batch training with large, static datasets becomes impractical. Instead, there is a need for learning algorithms that can continuously adapt and update in real-time while learning from human operators. This need forms the basis and objectives of the current study, as discussed in Sect. 1.

## Background

### Imitation learning in human-centric systems

Imitation Learning (IL) is a machine learning technique designed for learning directly from human behavior (Hussein et al., 2017). Instead of defining explicit rules or reward functions, IL enables a model to learn a task by observing and replicating demonstrations provided by human experts (Schaal, 1996). IL is particularly effective in domains where explicitly defining task objectives or reward functions is challenging, but expert behavior can be recorded. However, traditional IL often relies on static datasets, which limits its ability to adapt to new or unforeseen situations, leading to issues such as distributional shift, where the model encounters states not represented in the original demonstrations (Ross & Bagnell, 2010). A common solution to this problem is online learning to update learning policies continuously. In imitation learning, this involves keeping experts in the loop to provide ongoing guidance and new information. Active Learning improves data efficiency by involving humans in the labeling process more strategically. Instead of labeling all data, humans are asked to annotate only the



most informative samples selected by the model (Settles, 2011). Interactive Machine Learning goes a step further by making the human an active collaborator in the learning process. Humans are not limited to labeling data. Instead, they can provide real-time corrections and feature guidance, which allow adjustments to the model's behavior (Amershi et al., 2014).

Active Learning and Interactive Machine Learning primarily involve human input in the form of labels or feedback, but do not typically allow humans to modify the learning algorithm itself (Mosqueira-Rey et al., 2023).

### Dataset aggregation

A widely used interactive machine learning technique for addressing distributional shift in imitation learning is the Dataset Aggregation (DAgger) algorithm (Ross et al., 2011). DAgger reduces distributional shift by allowing the model to perform actions while a human expert provides corrective labels for the states it encounters. These expert-labeled instances are incrementally added to the training dataset, allowing the policy to adapt and improve iteratively. By combining autonomous execution with continuous expert guidance, DAgger demonstrates how interactive learning can lead to more robust and adaptive models in dynamic, real-world environments.

The data-gathering process in DAgger employs a probability-based strategy that alternates between expert and novice behaviors (Eq. 1). Here,  $\beta \in [0, 1]$  represents the probability that the system will choose the expert policy ( $\pi_{exp}$ ) or the novice policy ( $\pi$ ) when adding to the new training dataset. The meta-algorithm for determining the parameter  $\beta$  can follow various statistical strategies, such as weight decay, where the probability of using expert policies is initially high but gradually shifts towards novice policies as the novice's accuracy improves.

$$\pi_i(X_t) = \beta_i \pi_{exp}(X_t) + (1 - \beta_i) \hat{\pi}_{N_{i-1}}(X_t) \quad (1)$$

When  $\beta = 1$ , the system exclusively uses expert policies, while  $\beta = 0$  indicates that novice policies have full control of the system. In practice, the system follows a predefined schedule for  $\beta$ , where experts provide guidance without having complete control over the system.

### Human-gated dataset aggregation

Human Gated-DAgger (HG-DAgger) introduces a new implementation model that underscores the critical role of human operators in safeguarding the system (Equation 2). The gating function  $g(X_t)$  is designed not just to collect

observations from experts but also to allow the transfer of full control to the experts when necessary.

$$\pi_i(X_t) = g(X_t) \pi_{exp}(X_t) + (1 - g(X_t)) \hat{\pi}_{N_{i-1}}(X_t) \quad (2)$$

### Learning from multiple experts

Learning from multiple experts has been extensively studied, particularly in the context of offline learning, where the entire dataset is available from the beginning. In these settings, the learner can optimize its training process with full knowledge of the data. In contrast, online learning, where data arrives sequentially, presents additional challenges. One key difficulty is designing a no-regret algorithm, which aims to perform nearly as well as the best fixed strategy in hindsight, even without prior access to all the data. Algorithms like DAgger must carefully evaluate the reliability of expert feedback against predefined safety criteria to avoid accumulating poor decisions over time. To address this, Sun et al. proposed Multiple-Expert-Gated DAgger (MEGA-DAgger), which uses a Control Barrier Function as a gating mechanism to enforce safety constraints (Sun et al., 2024). Expert observations that violate these safety conditions are excluded from the aggregated dataset, preventing unsafe behaviors from being incorporated into the training process.

The approach used in MEGA-DAgger emphasizes the safety of critical systems but overlooks the nuances of human operators' behaviors. Operational styles are far more varied and complex. A less assertive operational style may not be critical to system safety but could result in suboptimal production outcomes, such as longer machining times. While these observations are less valuable than those from more attentive or skillful operators, they still contribute valuable data to the learner model. However, the contributions from different operators should not be treated equally; instead, they should be weighted differently based on their impact and relevance to the learning process.

### Autonomic computing

Autonomic computing, introduced by IBM in 2001, was proposed as a framework for developing self-managing and self-adaptive computer systems (Corporation, 2001). This paradigm draws inspiration from biological organisms, particularly the autonomic nervous system, which operates autonomously to manage and adapt to environmental changes. For instance, essential functions like heartbeat regulation and sweating are controlled seamlessly without conscious recognition or effort on the part of humans. IBM envisioned computer systems achieving a similar level of

autonomy, capable of self-governance and adaptation within their own environments.

To realize this vision, IBM proposed a control loop composed of five key building blocks: Monitoring, Analysing, Planning, Execution, and Knowledge. IBM's MAPE-K model and their vision of autonomic computing systems did not follow the conceptual model of machine consciousness, thus it does not differentiate between conscious and unconscious processes. Previous studies have proposed incorporating learning processes, such as reinforcement learning, into the MAPE-K framework. For instance, Tesauro suggested integrating reinforcement learning as a core component of the MAPE loop, enhancing each phase with a dynamic, learning-based approach to self-management (Tesauro, 2007). However, this approach constrained the MAPE loop to address specific functionalities within the system, rather than integrating knowledge and information across multiple processes or functionalities.

1. Monitoring-block collects information from external sources, such as sensors, and internal sources, including reactive processes, to provide a comprehensive view of the system's state.
2. Analysing-block processes the data received from the Monitor Block to generate actionable insights and a deeper understanding of the environment.
3. Planning-block uses the knowledge constructed by the Analyze Block and its own understanding of the environment to develop rational strategies, guiding the system toward optimal performance.
4. Execution-block implements the strategies devised by the Plan Block by disseminating information to the appropriate sub-processes for execution.
5. Knowledge-blocks serves as a repository for storing information, enabling future iterations of the MAPE-K loop to leverage past experiences. It facilitates the creation of new knowledge, which influences other blocks to adapt and refine their behavior over time.

Machine learning has been introduced several times in the past. One of the earliest attempts was to transform the entire MAPE-K loop into a reinforcement learning loop (Tesauro, 2007). Machine learning is commonly used to enhance various components of the MAPE-K loop in self-adaptive systems (Gheibi et al., 2021). It supports monitoring by improving data interpretation, analysis through prediction and anomaly detection, planning by optimizing adaptation strategies, and, in rare cases, execution by selecting appropriate actions. The choice of where ML is applied depends on the system's goals and the nature of the adaptation problem.

## Machine consciousness

Computational neuroscience and artificial intelligence (AI) have long been closely connected, sharing a common goal of understanding and replicating learning processes observed in the human brain. Computational neuroscientists aim to decipher how the brain processes information, uncovering the mechanisms behind biological learning and memory. AI researchers then apply these insights to develop effective algorithms. For instance, convolutional neural networks (CNNs) for vision tasks (Fukushima, 1980), Lecun et al. (1998) and recurrent neural networks (RNNs) for sequential processing (Hochreiter & Schmidhuber, 1997) draw inspiration from biological neural models, achieving remarkable success in their respective domains (Dong et al., 2021). By exploring machine consciousness within cognitive neuroscience, we aim to deepen our understanding of human learning processes, which can, in turn, be translated into advanced machine learning algorithms that foster a more nuanced model of human-augmented learning.

## Cognitive processes

The definition of consciousness remains a topic of debate, as different fields interpret it uniquely from their own perspectives. In computational neuroscience, one widely accepted definition involves the Global Workspace Theory (GWT) proposed by Baars (1988). GWT is a framework in cognitive science for understanding consciousness. It suggests that consciousness emerges when information is broadcast across a centralized "global workspace" in the brain. This workspace enables the integration of information from various unconscious processes—such as perception, memory, and reasoning, making it simultaneously accessible to multiple brain regions. Under this framework, information processing is divided into two categories: unconscious and conscious processing.

- *Unconscious Processing* Many complex mental tasks, such as recognizing faces or understanding speech, are carried out automatically by the brain without conscious awareness. When the output of such a process becomes strong or relevant enough, it crosses a threshold and enters the global workspace. This transition transforms the previously unconscious activity into a conscious process, allowing it to be accessed and used by other cognitive functions (Dehaene et al., 2003), Mashour et al. (2020).
- *Conscious Processing* Information is considered conscious when it becomes *globally accessible*—that is when it can be shared across various cognitive systems such as memory, attention, language, and decision-making.

Conscious processing typically involves active regulation, requiring deliberate attention, focused effort, or intentional choices beyond automatic responses. Equally essential is the second dimension of consciousness: *self-awareness* (Dehaene et al., 2017). This allows us to monitor and reflect on our own thoughts, recall past experiences, and engage in higher-level reasoning and complex problem-solving.

Building upon GWT, the Global Neuronal Workspace (GNW) model offers a more neuroscience-grounded perspective, suggesting that certain brain regions, particularly the prefrontal and parietal cortices, form a network that facilitates conscious processing (Dehaene et al., 1998), Mashour et al. (2020). The GNW highlights how specific brain areas act as hubs for integrating sensory and cognitive information, leading to the emergence of conscious awareness (Fig. 3). This concept has significant implications for AI research, as it provides a model for creating systems capable of managing and integrating vast amounts of information while prioritizing relevant data, ultimately moving AI closer to more human-like awareness and cognitive flexibility.

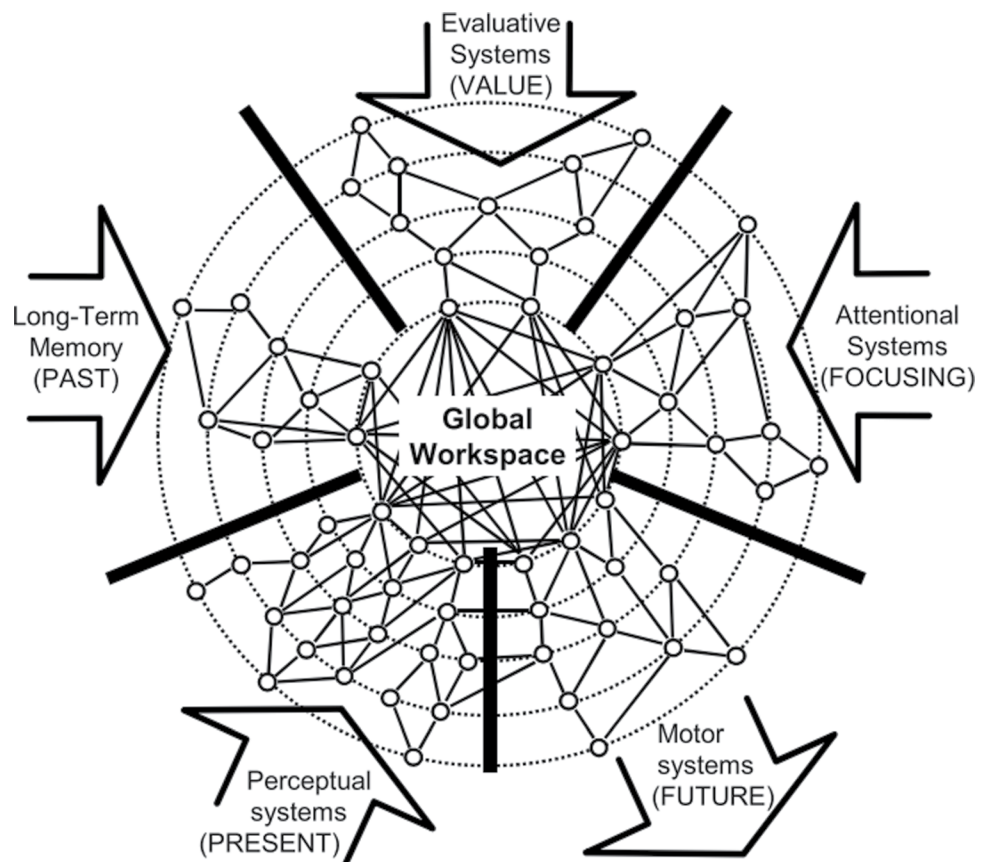
## Learning theory

Learning is the process of acquiring new knowledge through experiences. Evidence of learning is typically demonstrated by a change in behavior. However, behavior changes can be caused by other factors, like system disruption. To differentiate, the learning process is specified as changes in behavior through experiences (Lefrancois, 2019). Researchers from different disciplines might have different theories for learning. Still, they all agree that learning is incremental and iterative, which means that knowledge is gathered and accumulated via repeating cycles.

The learning process happens at both unconscious and conscious levels, each approaches in different ways to acquire and refine knowledge and skills.

- In the unconscious level, learning is characterized as an automatic and background process. The brain passively absorbs patterns, updating its internal models without conscious intervention. Some of the automatic learning models that have been agreed upon across different disciplines include subliminal priming (Kouider & Dehaene, 2007), reinforcement learning (Thorndike, 1933), and latent learning (Lefrancois, 2019), Tolman and Honzik (1930).

**Fig. 3** Global Neuronal Workspace representation with five main types of processors (Dehaene et al., 1998)



- Conscious learning reflects active, intentional learning. The process requires focus, attention, and mental effort (Dehaene et al., 2017). Active learning is an integral part of consciousness. It requires global accessibility of knowledge, information filtering, and memory recall. By reflecting on the information via abstract thinking and complex reasoning, it facilitates the creation of new knowledge, more nuanced and deeper. Conscious learning also includes meta-cognitive reflection allows our own learning process to identify errors, adjust strategies, and refine our approach based on this self-assessment.

The majority of human knowledge operates at the unconscious level (Baars, 1997), making it inherently difficult to formalize or articulate explicitly. This form of knowledge was first described as “tacit knowledge” by Michael Polanyi in 1966 (Polanyi, 1966). According to Baars, while actual encoding and processing of patterns happens implicitly, conscious awareness is required to trigger learning and the selection of learning materials (Baars, 1997).

The reinforcement learning mechanism in machine learning is the direct inspiration of its counterpart in biology. In this paradigm, agents interact with an environment, making decisions to maximize cumulative rewards over time (Sutton & Barto, 1998). Bayesian learning is a computational framework based on Bayes’ Theorem, which provides a principled way to update beliefs in light of new evidence (MacKay, 2002). Though not a direct model of biological learning, Bayesian learning closely aligns with cognitive theories suggesting that humans and animals often make decisions by weighing probabilities and updating expectations based on experience (Tenenbaum et al., 2006). These learning algorithms, however, focus on the unconscious learning process, facilitating an automatic, rigid framework to refine the established cognitive models rather than constructing more nuanced knowledge through reasoning or introspection.

## Augmentor

Building on theoretical frameworks of machine consciousness (Sect. 3.3), we align the first objective with reactive learning (Sect. 4.1), where the system passively refines its behavior through continuous observation and operator adjustments without explicit awareness. This process resembles how the human brain unconsciously absorbs patterns and regularities from the environment, automatically updating internal models. In contrast, the second objective aligns with context-aware adaptive learning (Sect. 4.2), where the system incorporates new insights and a deeper understanding of its environment. Expert input actively

shapes the learning trajectory, enabling the system to adapt more effectively and make informed decisions.

## Reactive learning

This section addresses *Objective 1* by introducing a continuous learning algorithm. The algorithm incrementally refines the behavioral cloning models developed in our previous work, where models were trained on a fixed dataset (Nguyen et al., 2023). This approach allows refinement by observing expert behavior in real-time.

**Definition 1** *Reactive learning* refers to a continuous, data-driven process in which learning algorithms automatically update their parameters based on incoming data, triggered by predefined conditions.

DAGger and other online machine learning algorithms are examples of reactive learning algorithms because they respond to predefined triggers. These triggers can be condition-based, such as a confidence threshold in DAGger, which reflects how closely the algorithm’s predictions match the expert policy. Alternatively, reactive learning can also be continuous, where the trigger is a predefined number of iterations or the sampling strategy. In DAGger, this involves alternating between the expert policy and the current learned policy, with learning triggered periodically as the system collects new samples according to the implementation strategy. In manufacturing environments, data is gathered from a broad range of experts rather than a select few. This approach offers the benefit of capturing a diverse array of experiences and scenarios, making data collection easier and reducing the risk of distributional shifts. However, a significant challenge arises from the variability in operational styles among different operators, shaped by their personal preferences and production needs. For instance, within the same production environment, some operators might adopt a cautious approach, slowing down machine speeds in advance to minimize risks, while others may take a more active approach, making real-time adjustments based on situational demands. Each style has its own advantages and drawbacks. The cautious approach helps preserve tools, reduces workload, and provides peace of mind for operators but at the cost of slower production speeds. On the other hand, the active approach optimizes production speed and maintains tool efficiency but demands greater attention and skill from operators. These human factors significantly influence the data collected, presenting challenges for machine learning algorithms to discern optimal patterns from the varied evidence available.

We propose a more nuanced version of Human-Gated DAGger that accounts for these variations by incorporating



them into the learner model through an instance weighting learning algorithm. In standard DAgger, the objective is to minimize the loss function over the aggregated dataset

$$\min_{\pi} \sum_{(x_t, a_t) \in \mathcal{D}} \ell(\pi(x_t), a_t) \quad (3)$$

where  $\ell(\pi(x_t), a_t)$  is the loss incurred by the policy  $\pi$  when it chooses action  $\pi(x_t)$  at state  $x_t$  compared to the expert action  $a_t$ . When incorporating instance weighting, the loss function is modified to weigh each instance according to the experts' importance.

$$\min_{\pi} \sum_{(x_t, a_t) \in \mathcal{D}} w(x_t, a_t) \cdot \ell(\pi(x_t), a_t) \quad (4)$$

Experts play a pivotal role in guiding the learning algorithm and intervening when necessary. The learning algorithm is assigned a confidence score, denoted as  $C$ . When experts do not intervene,  $C$  is high, indicating strong confidence in the algorithm's decision-making. Conversely, when experts intervene, the learning algorithm evaluates its decision-making by comparing it to the experts' behavior. In this case, the confidence score  $C$  becomes inversely proportional to the error rate between the algorithm's decisions and the experts' actions. From our experiments we found that mini-batch learning achieves faster convergence compared to stochastic learning. Instead of relying on a predefined threshold to trigger the learning process, we incorporate samples into the mini-batch for training based on the confidence score  $C$ . This approach ensures a more efficient and adaptive learning process. We adopt simulated annealing to control the inclusion of learned policy actions into the training dataset. It is to balance exploration—accepting learned policy actions, and exploitation—relying on expert corrections. The idea is to start by heavily favoring the expert's actions and gradually increase the probability of including learned policy actions as the training progresses, guided by a simulated annealing schedule. If the expert does not correct the learned policy's action, it means the learned action is already correct in that context. Including the learned policy's action in the dataset under these circumstances ensures that the policy remains consistent with its previously learned behaviors. This can help mitigate the risk of overfitting to changes in the expert's policy or potential inconsistencies in expert demonstrations. The probability of accepting a sample with a high confidence score  $C$  should depend on both the current temperature  $T$  and the confidence score:

$$P_{\text{accept}}(C) = \exp\left(-\frac{C}{T}\right)$$

In the early stages, we set a high temperature of accepting both high and low-confidence samples to allow the model to explore different possibilities and gather diverse data. As the model becomes more stable, we reduce the probability of accepting high-confidence samples, but still always accept low-confidence samples to allow the model to learn from uncertain or difficult cases. A cooling schedule is used to decrease temperature  $T$  over time to reflect growing stability in the training progress:

$$T_{\text{new}} = T_{\text{old}} \times \alpha$$

$\alpha$  is a decay factor that gradually reduces the temperature over time. This cooling schedule will guide how the system shifts from a more exploratory phase to a more selective phase.

### Context-aware adaptive learning

To achieve *Objective 2* of incorporating contextual information, environmental understanding, and operator knowledge into the learning system, we adapt the Global Workspace Theory and conscious learning theory to guide our learning algorithms.

Our enhanced HG-Dagger algorithm corresponds to a reactive learning process. The result of the process still needs a global workspace to allow broadcast to different computational processes and to be influenced by other processes in return. In the context of the Augmentor system, the two dimensions of consciousness can be reinterpreted as follows:

1. **Global accessibility:** a common knowledge representation that all computational processes must adhere to before their information can enter a global workspace. The global workspace functions as a centralized framework with defined rules that manage how knowledge is received, selectively focused upon, and broadcast to other computational processes. This enables the system to prioritize and integrate information, akin to selective attention in humans, where relevant details are emphasized while irrelevant ones are suppressed.
2. **Self-Awareness:** is represented by a rule set that enables self-reflection and the generation of new knowledge. This involves the system's ability to monitor its processes, evaluate past actions, and adapt to new challenges through iterative learning and reasoning. Unlike its biological counterpart, self-awareness in computational systems does not aim to replicate human introspection directly but rather serves as a functional abstraction to facilitate adaptive decision-making and complex problem-solving.

**Definition 2** *Context-aware adaptive learning* involves global accessibility of knowledge, integrating both reactive learning and contextual information. It requires self-reflective properties to derive new knowledge that actively enhances the reactive learning process.

We leverage the conceptual model of autonomic computing to design our computational global workspace. While Baars (1988) provided a theoretical framework explaining brain function through the division of unconscious and sub-conscious processes, IBM's autonomic computing can be utilized to translate the concept into a computational framework and outline how such systems could be realized in computer architecture. We build upon these foundations by integrating concepts from machine consciousness with technical frameworks from autonomic computing and machine learning to create a more nuanced version of the MAPE-K loop. In our design, computational processes are classified as follows:

1. *Reactive processes* These include learning models driven by data, such as online learning, pattern recognition, and other data-driven mechanisms.
2. *Context-aware adaptive Processes* These occur within the MAPE-K loop and actively influence reactive processes through the Execute block, ensuring adaptive and context-aware interactions.

Policies in the system can be created through two approaches: human-defined policies, where operators inject explicit policies based on their expertise and operational knowledge to guide system adaptation, and synthesized policies, where the system autonomously generates policies by analyzing accumulated evidence and contextual data, enhancing reactive processes with learned adaptations.

The MAPE-K in the Augmentor system becomes the global workspace for context-aware adaptive processes. The MAPE-K loop itself provides a mechanism for global accessibility between reactive processes. While the Analyse and Plan block utilizes information from the Knowledge storage to achieve the self-awareness dimension of machine consciousness (Fig. 4). SPARQL and semantic web technologies are used to implement adaptation policies within MAPE-K. Unlike previous approaches to integrating machine learning into MAPE-K (Gheibi et al., 2021), our design treats learning algorithms as distinct reactive processes that report their outputs to the global workspace component of the MAPE-K loop. This separation of concerns enables modular scalability, enabling the integration of multiple models and processes seamlessly.

Table 2 presents a comparative summary of context-aware adaptive process learning alongside various DAGger

variants commonly used in active learning. Unlike traditional approaches that rely solely on expert demonstrations, Augmentor enables adaptive learning through contextual reasoning, elevating the expert's role from a mere label provider to a source of semantic and procedural knowledge. This shift transforms the expert's involvement from simply supplying labels to contributing richer semantic and procedural input. Additionally, Augmentor introduces a mechanism for sharing information across different learning processes through a global workspace, allowing multiple learners to benefit from each other's experiences—resembling the learning systems found in human cognition.

## Experiment

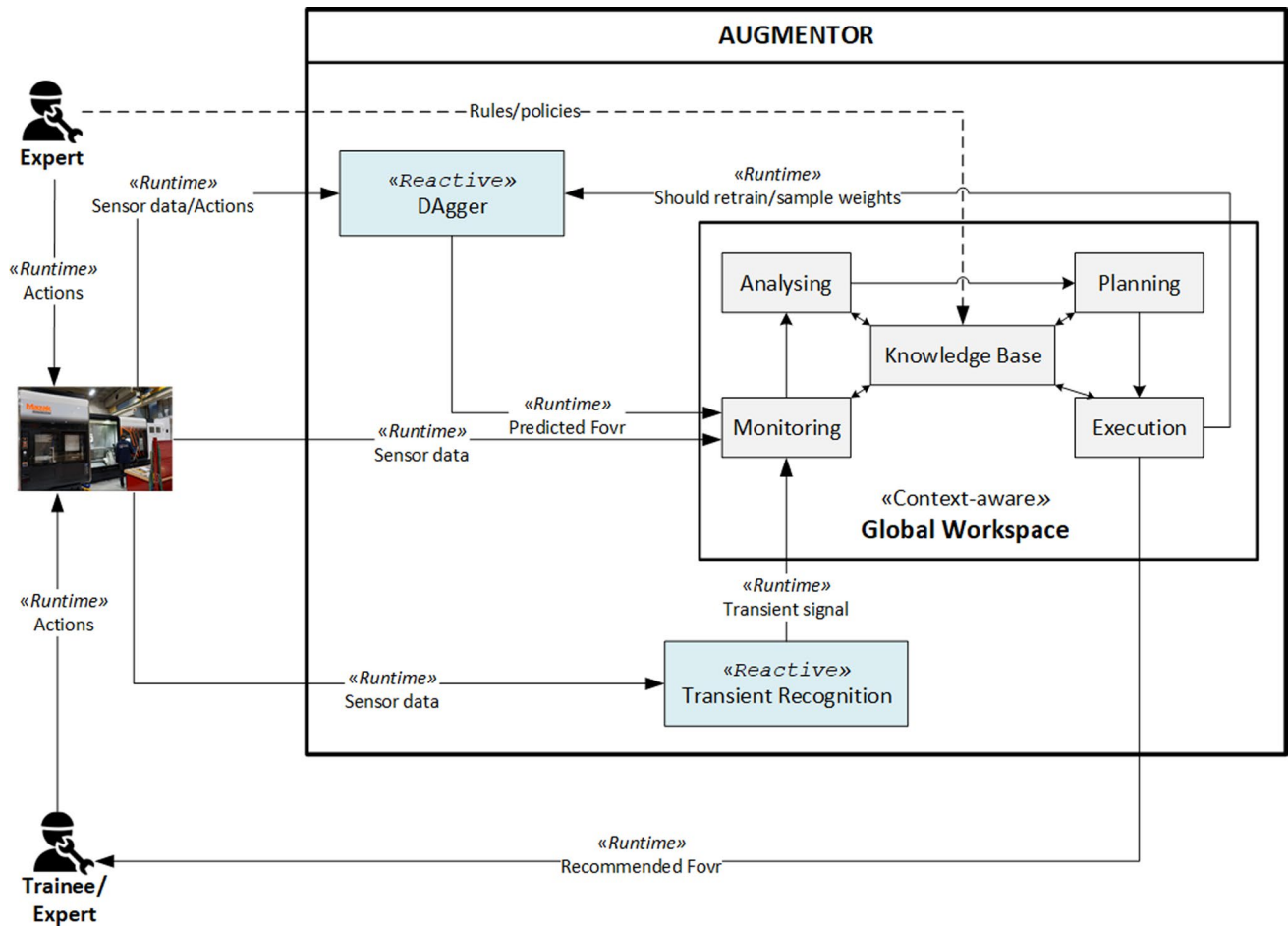
The experiment was designed to validate the two main objectives of developing a context-aware adaptive learning mechanism.

### Data preparation

For our experiment, we collected data from the Mazak Integrex i500 at MSH over four months, spanning October 2023 to January 2024. During this period, the machine produced multiple product series.

**Definition 3** An *episode* refers to the period during which the CNC machine produces a unit of the product design. This concept is adapted from reinforcement learning, where an episode represents a complete sequence of interactions between an agent and its environment, beginning in an initial state and concluding in a terminal state (Sutton & Barto, 1998).

A chuck is a tool that holds and rotates the workpiece during machining. The machine operates only when the chuck is in the CLOSED state. We used the *ChuckState* signal to divide the sensor data stream into separate episodes. When the *ChuckState* transitions from OPEN to CLOSED, it marks the start of an episode. Conversely, when the *ChuckState* transitions from CLOSED to OPENED, it signifies the end of the episode, as the operator removes the finished part from the chuck grip. Episodes are grouped into a series if they share the same sequence of tool numbers and their timestamps are within a close interval, within one day or one week.



**Fig. 4** Overview architecture of Augmentor, which serves as a knowledge preservation tool by learning from experts through both reactive learning and expert-guided input. The system passively refines its behavior through continuous observation and operator adjustments (Reactive processes), while actively incorporating new insights and

contextual understanding to enhance decision-making and performance (Context-aware Global Workspace). Knowledge gained can be transferred to other trainees or experts in the form of actionable recommendations

**Table 2** Comparative summary of different learning algorithms

|                               | Dagger                      | HG-Dagger                             | MEGA-Dagger                                     | Augmentor                                                                      |
|-------------------------------|-----------------------------|---------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------|
| Goal                          | Reduce compound-ing errors  | Improve safety via gating             | Filter unsafe/conflicting data                  | Adapt using context and semantics                                              |
| Learning source               | Expert demonstra-tions only | Expert demonstrations gated by safety | Filtered input from mul-tiple imperfect experts | Combines expert actions with contex-tual and domain knowledge                  |
| Expert role                   | Full demonstrator           | Safety gate                           | Imperfect label provider                        | Context source and adaptive guide                                              |
| Context use                   | None                        | Partial (safety only)                 | Scenario-specific                               | Deep, semantic, task-aware                                                     |
| External knowledge use        | None                        | Minimal (safety judgment)             | Task-specific heuristics                        | Actively integrates semantic, environ-mental, and task-level context           |
| Cross-agent knowledge sharing | None                        | None                                  | None                                            | Multiple learners influence each other via shared context and global workspace |

**Definition 4** A *series* is a collection of episodes in which the CNC machine produces similar parts based on the same design.

For each episode, we measure the production time for during which the *Execution* status is ACTIVE. Episodes with shorter production times are assumed to be of higher

value. Feedback from operators for these episodes is given greater weight compared to others. Table 3 lists the series selected for our experiment. The selected series includes at least four episodes and involves the use of at least four different cutting tools. The benefits of organizing data sequences into episodes and series include:

**Table 3** List episodes used for experiment, duration and active time in minutes

| N. Eps | Duration | Active time | N. Tools |
|--------|----------|-------------|----------|
| 11     | 22.62    | 12.95       | 8        |
| 6      | 73.56    | 46.98       | 7        |
| 7      | 171.67   | 133.34      | 10       |
| 9      | 30.16    | 22.26       | 6        |
| 5      | 18.85    | 8.91        | 9        |
| 5      | 24.64    | 15.31       | 5        |
| 5      | 8.35     | 7.63        | 6        |
| 16     | 41.68    | 31.72       | 8        |
| 17     | 64.94    | 53.45       | 5        |
| 11     | 19.20    | 10.94       | 6        |
| 12     | 36.23    | 13.59       | 4        |
| 11     | 19.64    | 6.62        | 4        |

1. By separating the data into episodes, we can concentrate on meaningful periods when the machine is actively working, effectively eliminating idle times such as breaks or pauses.
2. Typically, when operators begin working on a new design or series, the initial episodes are longer and more error-prone as operators adapt their strategies and refine their approach to the new requirements. This separation helps identify and analyze these learning phases.
3. Different series often come with varying requirements, leading to variations in the quality and type of skills operators invest in. By grouping data into series, we can better understand these differences and their impact on production outcomes.

## Emulation

We conducted our experiment using emulation instead of real-time data for the following reasons:

1. The CNC machine at MSH is an industrial machine, and its scheduling is beyond our control, being dependent on the ongoing business operations at MSH.
2. Machining times vary significantly depending on the specific requirements. In some cases, the machining process can be extremely slow to ensure the highest quality of the products, which may not align with the timelines of our experiment.
3. The CNC machine is expensive and operates under strict rules. Since our learning algorithms are still in the experimental phase, direct testing on the machine poses a significant risk of causing damage to the machine, the products, and potentially the human operators.

We developed an emulation system that replays the recorded data stream, enabling us to conduct experiments at an accelerated pace. This approach ensures the safety of

the CNC machine, products, and operators while providing a controlled environment for testing and validating our algorithms.

For the experiment, we assume that operators observe the current data, there will be a small time gap for their reaction in their reactions.  $Action_{t+g} = f(State_t)$ . Emulation replay data as recorded. Augmentor will use current replaying data to make a prediction of what would be the appropriate reaction after  $g$  time  $Pred_{t+g} = Augmentor(State_t)$ . The emulation then compares the  $Action_{t+g}$  and  $Pred_{t+g}$ .

Comparing  $Action_{t+g}$  and  $Pred_{t+g}$ , if the error margin is smaller than *threshold*, we assume that the operators follow the predictions from Augmentor. If the error margin is bigger than *threshold*, we assume that operators intervene and inject their knowledge about the situations. Augmentor will then try to update the models.

The confidence score  $C$  of  $Pred_{t+g}$  is calculated:

$$C = 1 - \text{normalized}(\text{abs}(Pred_{t+g} - Action_{t+g}))$$

which gives the absolute confidence of 1 when the prediction matches the action from the expert.

## Evaluation metric

Regret is a key concept used to evaluate the performance of online learning algorithms (Ross et al., 2011). It is defined as the difference between the average loss incurred by the sequence of learned policies and the average loss incurred by the best policy in hindsight.

$$Regret = \sum_{i=1}^N l_i(\pi_i) - \min_{\pi \in \Pi} \sum_{i=1}^N l_i^*(\pi) \quad (5)$$

where

- $l(\pi(x_t))$  is the loss incurred by the policy  $\pi$  learned by the DAgger algorithm in iteration  $i$ .
- $\pi^*$  is the expert policy or the best policy in hindsight that minimizes the cumulative loss
- $N$  is the total number of iterations or time steps

A no-regret algorithm is one for which the average regret converges to zero as the number of iterations grows (Ross et al., 2011).

$$Regret_N = \frac{1}{N} Regret \leq \gamma_N \text{ for } \lim_{N \rightarrow \infty} \gamma_N = 0$$

Regret quantifies the performance gap due to making decisions without knowing future data (Ross et al., 2011). If the



learner has high regret, its policies are significantly worse than the expert policy, and it has not learned effectively from the expert's corrections. Low regret indicates that the learner is performing close to the expert's level and is effectively learning from feedback, approaching expert-level performance over time. In Eq. (5), regret is measured against a single fixed action that is optimal in hindsight. This limits its ability to capture the adaptability of an algorithm in changing settings. Dynamic regret allows evaluation against a sequence of comparators that can vary over time, making it better suited for dynamic environments (Zhang et al., 2018).

$$\text{Regret} = \sum_{i=1}^N l_i(\pi_i) - \sum_{i=1}^N l_i(\pi_i^*)$$

In the context of DAGger, the expert policy at each timestep,  $\pi_i^*$ , is assumed to produce the optimal action. Consequently, the expert's loss is typically considered zero or negligible when making comparisons.

$$\text{Regret}_N = \frac{1}{N} \sum_{i=1}^N l_i(\pi_i)$$

The loss  $l(\pi_t)$  for the learner policy  $\pi_t$  at time  $t$  is typically computed using the chosen metric. In our case, we use mean squared error (MSE) between the actions taken by the learner and the actions taken by the expert on a dataset of states  $s_t$ :

$$l_i(\pi_i) = \frac{1}{m} \sum_{j=1}^m (\pi_i(x_j) - a_j)^2$$

where  $\pi_i(x_j)$  is the action taken by the learner policy  $\pi_t$  on state  $x_j$ , and  $a_j$  is the action taken by the expert on the same state, and  $m$  is the total number of samples.

One limitation of using regret as a performance measure is that its interpretability can become less straightforward when dealing with subtle differences across large datasets or multiple objectives (Rakhlin et al., 2011). In this study, the data were broken down into modular levels, series, and episodes, allowing for closer inspection of how different conditions or phases affect learning.

## Experimental procedure

In our experiment, we used the emulator to replay the episodes listed in Table 3 sequentially within the same series, following their chronological order. The series themselves were replayed in random order. Our learning system

received data from the emulator in real-time and processed it to learn control patterns. Three learning algorithms were executed in parallel within our learning system:

1. *Static model* was created from our previous work (Nguyen et al., 2023).
2. *Reactive models* build upon the static model as its initial foundation. It then continuously updates itself using our enhanced HG-Dagger algorithm, as described in Sect. 4.1. This process operates as a reactive learning mechanism.
3. *Context-aware adaptive models* follow the same procedure as the learning model but are monitored and regulated by the global workspace MAPE-K, as detailed in Sect. 4.2. This process represents a context-aware adaptive learning mechanism.

We measure regrets to evaluate the effectiveness of the learning models. Hypothetically, as our learning algorithms evolve through feedback from operators, we expect to observe the following outcomes:

1. How the learning happens across episodes in the same series: The extent to which the algorithms learn from earlier episodes compared to later episodes within the same series. This will help us assess how well the algorithms capture and leverage operator strategies as they refine their approach over time.
2. How the learning adapts across series: The level of adaptation required to transition from one series to another with different requirements. This observation will allow us to evaluate the algorithms' ability to generalize and effectively adapt to changes in production scenarios and requirements.

The comparison between reactive models and the static model demonstrates the proposed system's effectiveness with respect to *Objective 1*. Context-aware adaptive models address both objectives; therefore, comparing the context-aware adaptive model with the reactive model highlights the proposed system's effectiveness with respect to *Objective 2* (Sect. 1).

### Initial static model

In our previous work, the original dataset used for creating the static model was gathered over a month from the Mazak Integrex i500 (Nguyen et al., 2023). During this period, the CNC machine produced identical units of the same product. The maximum data frequency was set at 4Hz, with sensor data only updating upon changes. This occasionally led to inconsistent data frequencies. To standardize the data, it was

interpolated to a steady 4Hz frequency. The CNC machine could be paused during the machining process, such as during lunch breaks. To exclude these inactive intervals, we used the *Execution* signal to filter out the periods when *Execution* was not ACTIVE.

A cutting path is defined as the interval beginning when the cutting tools engage the workpiece and ending when they lift and rapidly advance to the next cutting position. During the movement to the next position, the load on the turning spindle drops almost to zero, while movement speed and feed rates spike briefly. These intervals are considered transient periods, which can increase the variance of our dataset. We removed these segments from the training dataset. Identifying sudden shifts in time series behavior is known as change point detection. For this, we applied the PELT (Pruned Exact Linear Time) algorithm from the ruptures library (Truong et al., 2020). PELT detects an unknown number of change points, assuming they occur linearly (Killick et al., 2012). Abrupt changes in spindle load can also result from irregular surfaces. To differentiate between cutting paths and transient periods, segments with a quantile below ten between change points are labeled as transient and discarded (Fig. 5)

While our previous study found that Random Forest performed similarly to Long Short-Term Memory (LSTM) on our offline dataset (Nguyen et al., 2023), deep neural networks (DNNs) tend to offer greater advantages when working with time series data, especially in the context of active learning. One key benefit is that DNNs can learn useful features directly from raw input without the need for manual feature engineering or transformation (Ismail Fawaz et al., 2019). This is particularly useful in situations where domain knowledge is hard to formalize, and where relying on data alone becomes necessary. Moreover, DNNs family can be trained more efficiently on GPUs, which makes them better suited for handling high-dimensional time series data in online learning settings (Ismail Fawaz et al., 2019). When

paired with smart sampling or data selection strategies, deep models can also deliver strong performance in active learning scenarios (Shen et al., 2017).

### Reactive learning processes

The learning model follows our enhanced HG-Dagger algorithm in Sect. 4.1. It starts with a high temperature  $T_0$  and a high probability of accepting both high and low-confidence samples. As temperature decreases, high-confidence samples are rarely accepted, but low-confidence samples are always accepted. The model focuses more on refining its understanding of the control patterns presented in the time series data. One key performance metric in CNC production is active time, with shorter durations being preferable. As a result, data from episodes with shorter active times are assigned higher sample weights compared to those with longer production times within the same series.

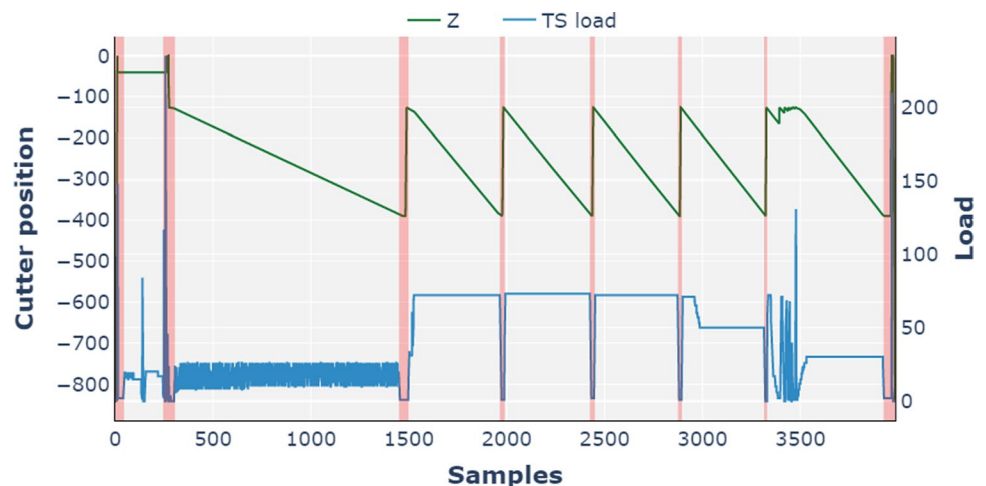
Our DAgger algorithm exhibits elements of subliminal priming learning. It passively absorbs information from stimuli when the mini-batch dataset becomes available for updating the model. In the absence of contextual input from the global workspace, DAgger operates according to its default algorithm, with sample weights maintaining their standard value of one.

### Context-aware adaptive learning processes

During the initial phase of generating the static model, we inadvertently integrated contextual knowledge into the learning process through data preprocessing to create the training dataset. However, two key contextual elements are inherently absent from the reactive learning process:

- The model does not perform updates using data collected when the machine is not in an ACTIVE state.

**Fig. 5** The green line visualizes the toolpath on Z-axis. Transient periods are highlighted as light pink when the cutter quickly moves to the next position for the cutting path



**Fig. 6** When the machine is not active, do not update the learning models

```
PREFIX aug: <http://example.com/aug#>

DELETE {
  aug:Mazak aug:shouldRetrain "false" .
}
INSERT {
  aug:Mazak aug:shouldRetrain "true" .
}
WHERE {
  aug:Mazak aug:hasExecutionState "ACTIVE" .
}
```

**Fig. 7** Adjust the sample weights based on how importance the experiences

```
PREFIX aug: <http://example.com/aug#>

DELETE {
  aug:Mazak aug:hasSampleWeight ?oldWeight .
}
INSERT {
  aug:Mazak aug:hasSampleWeight 0.5 .
}
WHERE {
  aug:Mazak aug:hasCluster "high" .
  OPTIONAL { aug:Mazak aug:hasSampleWeight
    ?oldWeight . }
};

DELETE {
  aug:Mazak aug:hasSampleWeight ?oldWeight .
}
INSERT {
  aug:Mazak aug:hasSampleWeight 1 .
}
WHERE {
  aug:Mazak aug:hasCluster "low" .
  OPTIONAL { aug:Mazak aug:hasSampleWeight
    ?oldWeight . }
}
```

- The model recognizes the variance introduced during transient periods and excludes such data from the training dataset.

In the Augmentor system, MAPE-K serves as the global workspace, functioning as the central hub for contextual information and active awareness.

- Context-awareness: we implemented a set of rules using SPARQL and semantic web technologies to enable contextual awareness, transforming reactive learning into context-aware adaptive learning. Figure 6 illustrates how the system informs the reactive learning process to identify idle periods when the machine is inactive and exclude such data from the learning process. Figure 7

demonstrates the system's ability to assign varying levels of importance to knowledge from different episodes. The importance of each episode is classified based on productive time, with shorter productive times indicating higher experimental quality, which in turn contributes to higher sample weights.

- Global Accessibility: in the initial creation of the static model, transient periods were identified using the PELT algorithm (Killick et al., 2012). It is an offline algorithm requiring the observation of the entire time series to detect change points. To address this limitation, we developed a separate reactive process for transient recognition using machine learning (Fig. 4). The PELT algorithm was employed to label samples as transient or not, creating a labeled dataset. This dataset was then used

**Fig. 8** Only update the models using samples that are not transient

```
PREFIX aug: <http://example.com/aug#>

DELETE {
  aug:Mazak aug:shouldRetrain "false" .
}
INSERT {
  aug:Mazak aug:shouldRetrain "true" .
}
WHERE {
  aug:Mazak aug:isTransient "true" .
}
```



**Fig. 9** Regret trajectories of different models. Light gray vertical lines indicate the division of time into episodes, while black vertical lines further subdivide the episodes into series

to train a pattern recognition model capable of identifying whether a given data sample is transient. We use Random Forest as the classifier for transient recognition (Liaw & Wiener, 2002). During the experiment, transient recognition operates as an independent reactive process, reporting its findings to the MAPE-K system. MAPE-K uses this information to determine and broadcast appropriate adaptation policies. It decides whether the learning process should proceed with updates based on the transient recognition results, as illustrated in Fig. 8.

## Results

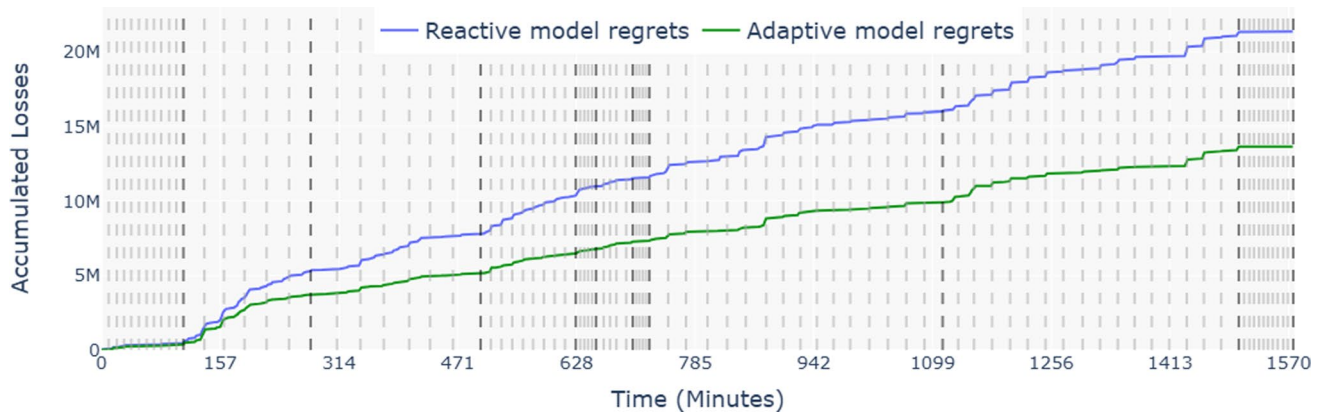
Figure 9 illustrates the accumulated regret trajectories for the static, reactive, and context-aware adaptive models. The static model exhibits the worst performance, with a steadily increasing regret trajectory of much higher magnitude than those of the learning models. This pattern suggests that the static model, trained on a limited set of real-life scenarios, fails to generalize and adapt, resulting in a persistently high error rate. In contrast, the learning-enabled models exhibit substantially flatter regret trajectories, indicating that the

ability to learn over time significantly improves decision-making performance and adaptability.

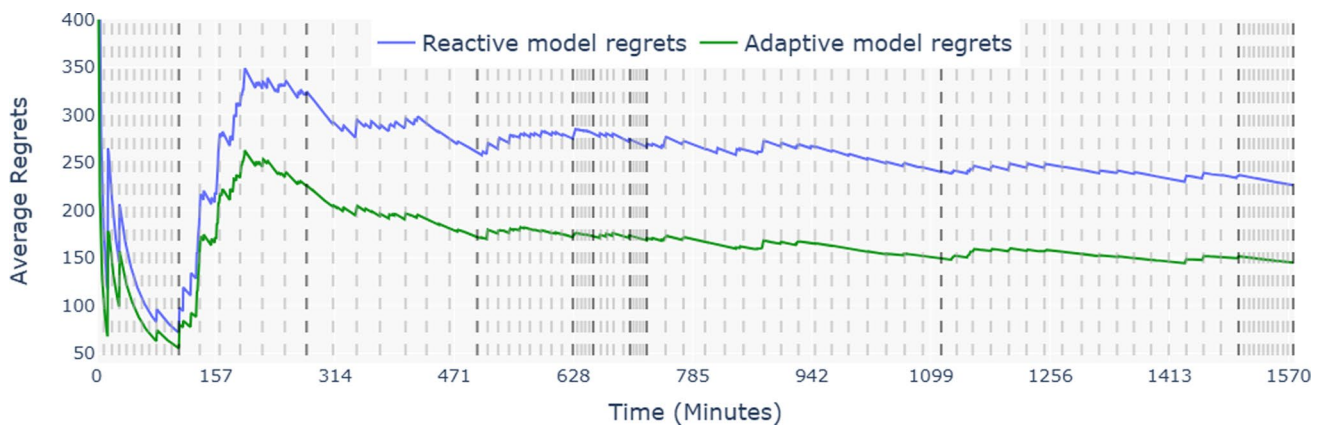
Figure 10 provides a zoomed-in view of the learning model trajectories, highlighting the benefit of incorporating context-awareness into the learning process. The adaptive model, which leverages contextual information, achieves a lower regret slope than the reactive model, reflecting more accurate and consistent decision-making over time.

Figure 11 presents the average regret trajectories of the learning models. Both models initially exhibit a sharp decrease in average regret after the first series, followed by a sudden increase, suggesting overfitting to the early episodes. However, both models eventually adapt, and their average regrets show a trend toward zero, consistent with the no-regret property. Moreover, the adaptive model demonstrates less volatility than the reactive model, indicating improved robustness and stability through context-awareness.

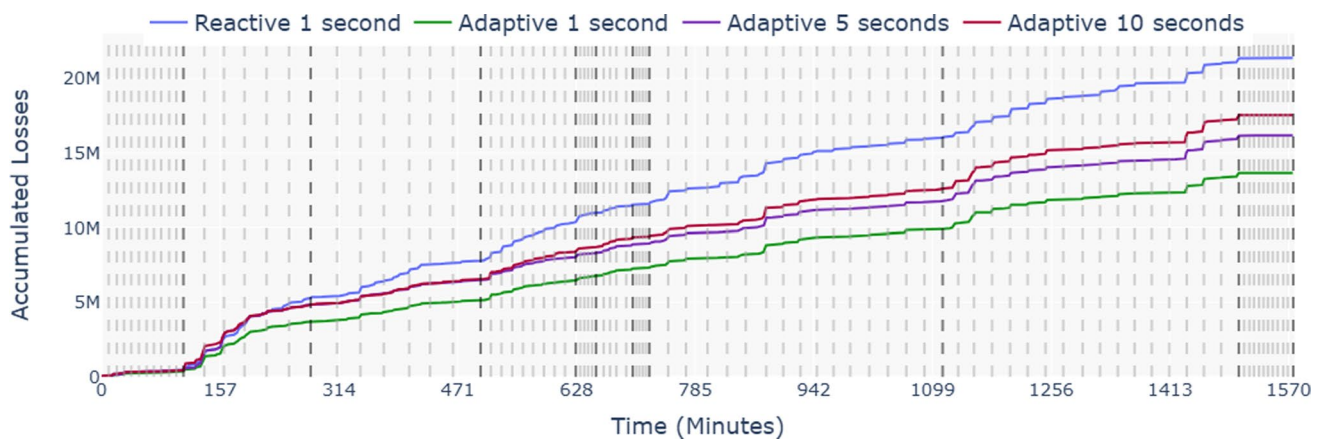
The light gray vertical lines represent the division of time into episodes, whereas the black vertical lines provide additional subdivisions within each episode, forming series (Figs. 9, 10, 11, 12, 13). Across episodes within the same series, a consistent trend emerges: regret initially increases sharply at the beginning of each series and then levels off in later episodes (Fig. 10). This pattern reflects the behavior of



**Fig. 10** The zoomed-in view of Fig. 9, focusing on the learning models. Light gray vertical lines indicate the division of time into episodes, while black vertical lines further subdivide the episodes into series



**Fig. 11** The average regret trajectories of the learning models. Light gray vertical lines indicate the division of time into episodes, while black vertical lines further subdivide the episodes into series



**Fig. 12** Prediction horizon of the reactive model with  $g = 1$  second, and informed models with  $g$  of 1, 5, and 10 s. Light gray vertical lines indicate the division of time into episodes, while black vertical lines further subdivide the episodes into series. The y-axis is plotted using a log scale

human operators as they gradually adapt their knowledge to a specific series. The result is a staircase-like trajectory, where regret spikes at the start of each new series, indicating that the current learning model struggles to generalize effectively across all scenarios. This trend is highlighted in

Fig. 13, which shows the growth rates of regret across episodes. Within a single series, growth rates are highest during the first few episodes and decrease significantly in later episodes. However, the slope of regret increase between series diminishes in later series compared to earlier ones,



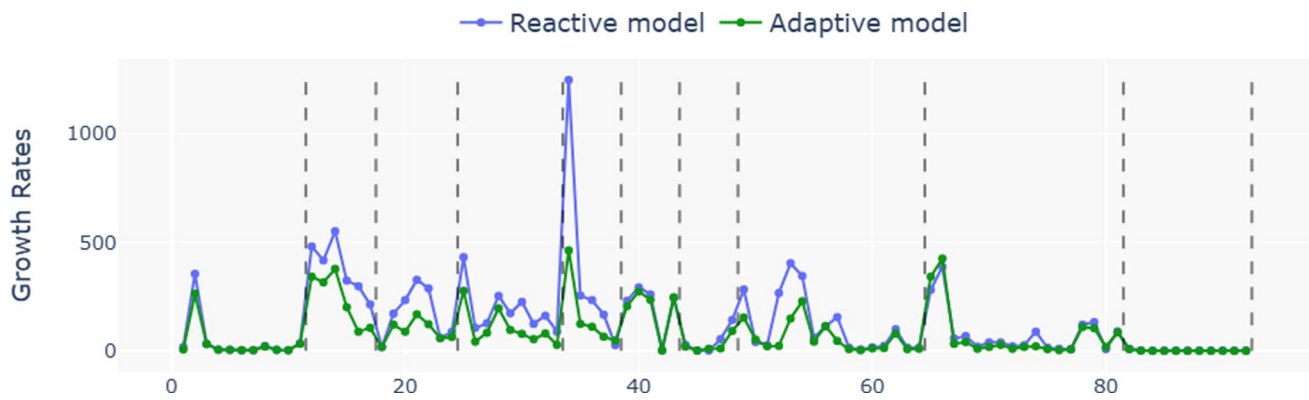


Fig. 13 Growth-rates of regrets over episodes. The dots indicate individual episodes. Black vertical lines divide the episodes into series

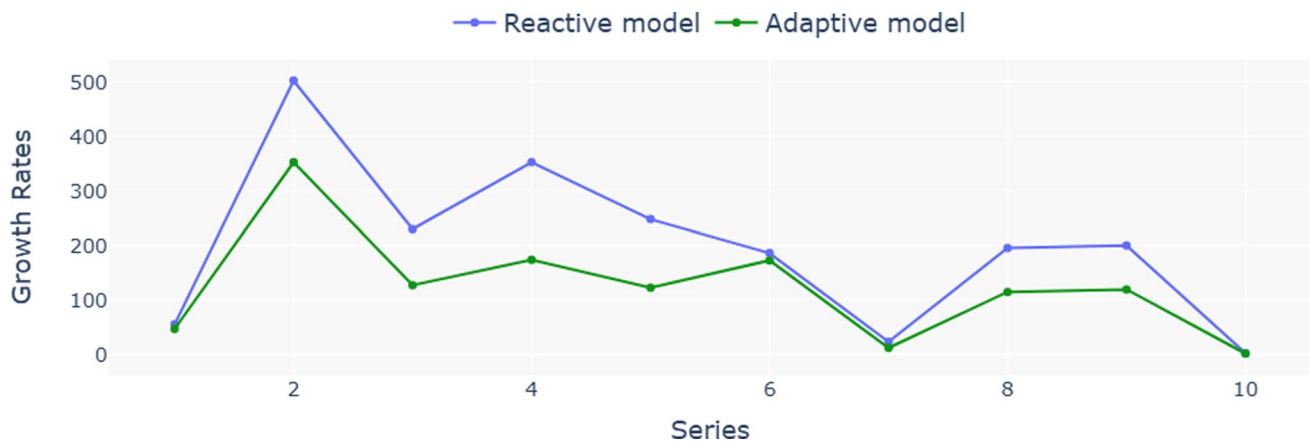


Fig. 14 Growth-rates of regrets over series. The dots indicate individual series

suggesting that the knowledge gained by the learning algorithm is progressively contributing to improved performance in subsequent series. Transitioning from one series to another requires adaptation, leading to a temporary increase in growth rates (Fig. 13). Nonetheless, Fig. 14 shows a broader downward trend in regret across series, with later series exhibiting lower regret growth rates compared to earlier ones. This demonstrates the learning model's gradual improvement in adapting to new series over time.

The key difference between the reactive learner and the adaptive learner lies in the incorporation of additional contextual knowledge, which introduces bias or information gain. This contextual knowledge can take the form of rules or policies derived from human understanding or inferences generated directly from the knowledge base.

$$Regret_{context\_aware} = Regret_{reactive} - bias$$

Figure 12 illustrates the impact of varying the time gap  $g$ , which determines how far into the future the model predicts. As the time gap  $g$  increases, accumulated losses, represented as regrets, also grow. This reflects a trade-off between prediction accuracy and the length of the prediction horizon.

While a longer time gap facilitates long-term planning, it reduces the model's prediction accuracy. However, when compared to the regret trajectory of the reactive model, the reactive model with a time gap of  $g = 1$  still exhibits significantly higher regret. This indicates poorer performance compared to the adaptive models with time gaps of  $g = 5$  and  $g = 10$ . This emphasizes the importance of knowledge enrichment in the learning process, which helps compensate for a longer prediction horizon. These findings demonstrate another practical application of knowledge enrichment—by increasing the prediction horizon, additional context information can be provided to offset the challenges of making further predictions.

## Discussion

### Regarding the main objectives

Objective 1 aimed to continuously refine the imitation learning algorithm by learning from operators' actions in real-time. This was achieved through the enhanced HG-Dagger algorithm, which incrementally updated behavioral cloning

models based on incoming data and expert feedback. The results show that the reactive learner, when compared to the static model, significantly reduced cumulative regret over time. This demonstrates that the model effectively assimilated operator behavior and refined its control policies through repeated exposure and feedback. The staircase pattern in regret across episodes further confirms this learning dynamic—each new series initially incurred high regret that gradually decreased as the model adapted, illustrating how learning accumulated and improved through experience.

Objective 2 focused on incorporating contextual knowledge, environmental understanding, and operator insight into the learning system. This was addressed through the context-aware adaptive learning framework, which augmented reactive learning with information broadcast through a centralized global workspace based on the MAPE-K architecture. Unlike the reactive model, the context-aware model was able to selectively filter irrelevant or noisy data during machine inactivity or transient states, and dynamically adjust sample weights based on contextual relevance. These context-sensitive adaptations led to a further reduction in regret compared to the reactive learner alone, validating the effectiveness of integrating high-level expert knowledge and reasoning into the learning loop.

Together, these results demonstrate that the Augmentor system meets both objectives—reactively learning from expert demonstrations and adaptively improving through structured, context-aware reasoning.

### Actionable insights

The study focused on a specific form of tacit knowledge involved in controlling feed rate. It showed how this type of knowledge can be modeled as a reactive, data-driven process within Augmentor which is functionally equivalent to unconscious learning in Baars's Global Workspace Theory (GWT) (Baars, 1988). In line with Baars' theory (Baars, 1997), the triggering of learning and the reception of relevant input through the global workspace reflect how conscious access guides what is learned, while the actual processing remains unconscious.

The experimental results not only validate the effectiveness of the proposed learning framework but also offer several insights that can inform the design of future adaptive systems:

- Inspired by the Global Workspace Theory, this study shows that learning processes can be organized as modular, reactive agents whose outputs are selectively broadcast and coordinated through a shared global workspace. This architecture facilitates learning across processes, enabling multimodal integration and information reuse,

and opens new possibilities for designing AI systems that mirror human cognitive structure.

- Traditional online learning systems treat all observations equally, often failing to account for environmental context or operational relevance. By incorporating a context filter and dynamic sample weighting, our approach allows learning algorithms to prioritize meaningful data, ignore irrelevant signals, and adapt more effectively. This breaks the rigidity of typical online updates and demonstrates a path toward more intelligent, context-sensitive learning.
- The experimental framework allows different learning processes to contribute and access shared knowledge in a coordinated way. This illustrates how multimodal learning—including signals from environment states, expert actions, and contextual rules—can be integrated through a central knowledge base, allowing the system to learn more efficiently and act with greater coherence.
- The enhanced Human-Gated DAGger algorithm underscores the importance of keeping humans in the loop—not simply as data sources, but as active collaborators in guiding and correcting the learning process. By embedding human feedback into the system's adaptation cycle, this cooperative model ensures that humans grow and adapt alongside the system, fostering a mutually reinforcing learning relationship. Rather than seeking to replace human involvement, the design encourages continuous co-evolution, where the system aligns with human goals while empowering users to remain engaged, informed, and in control.

### Challenges and future directions

This study concentrates on offline imitation learning. While it doesn't offer the flexibility of interactive, real-time approaches, it comes with clear benefits—especially in terms of safety, repeatability, and ease of deployment. These strengths make it well-suited for situations where live system access is limited or where safety concerns prevent direct experimentation. In future work, we plan to bring these algorithms into real-world deployment, allowing them to actively influence the system's behavior and adapt based on real-time feedback.

This study concentrated on one specific operator task: adjusting the override feed rate. In practice, however, CNC operators are responsible for a range of other activities, such as changing tools, clearing chips, and responding to unexpected events. These tasks often require operators to rely on multiple sources of information, including what they hear and see during machine operation. Expanding this work to better capture the full scope of operator behavior would therefore require incorporating multimodal data. Recent

developments in sensing hardware and multimodal large language models (MLLMs) (Zhang et al., 2024) make it increasingly feasible to gather and interpret data from various channels, such as sound, video, and temperature readings. While this study relied on manually defined SPARQL rules to represent contextual information, MLLMs offer a compelling alternative to the traditional knowledge base in the MAPE-K loop, which serves as the core of the global workspace. MLLMs have the potential to automatically derive contextual rules from unstructured sources such as maintenance logs and operator notes, while maintaining semantic alignment with sensor data and supporting consistent, context-aware reasoning. However, these models still lack reliable checking mechanism to ensure the accuracy and safety of the knowledge they generate (Chen & Shu, 2024). The primary challenge in implementing the MAPE-K framework as a global workspace lies in constructing the internal knowledge base, particularly in managing domain-specific knowledge, which still requires the guidance of domain experts.

Our industry partner, MSH, has pointed to the increasing shortage of skilled CNC operators as a pressing challenge. One proposed solution is to allow a single operator to supervise multiple machines at the same time. To support this kind of scalability, the system would need to adopt a multi-agent design, where each Augmentor unit can make decisions independently while still coordinating through a shared global workspace. Implementing this would involve complex scheduling and possibly planning strategies (Wooldridge, 2009) to manage decision-making across machines, or even across multiple factory locations. Furthermore, incorporating data on energy consumption and tool wear into the framework would make it more useful for addressing goals related to sustainability and cost control, which are becoming more important in modern manufacturing environments.

By addressing these challenges and leveraging the strengths of modularity, formal knowledge representation, and adaptive learning, the outlined framework holds significant potential for generalization across diverse and complex domains.

## Conclusion

In this paper, we introduced the Augmentor framework for context-aware adaptive learning, which enables the integration of contextual knowledge into online learning algorithms. Drawing on conceptual theories of machine consciousness, we categorized computational processes into two distinct types:

1. **Reactive processes:** These include data-driven algorithms such as learning models and pattern recognition techniques. Their primary function is to refine model parameters and adapt to new information embedded in the data.
2. **Context-aware adaptive processes:** These processes manage and integrate contextual information through the global workspace. They ensure global accessibility and facilitate the formation of new knowledge through human guidance and knowledge injection, enhancing the system's adaptability and decision-making.

We demonstrated the application of the Augmentor framework through an industrial use case focused on digitizing tacit knowledge from human operators. The system achieves two key objectives:

1. **Continuous updates to learning algorithms with new evidence from expert interventions:** To meet this objective, we proposed an enhanced version of the DAGger algorithm that incorporates global accessibility to the global workspace, facilitating the fulfillment of the second objective.
2. **Integration of contextual knowledge into the learning system:** Leveraging the autonomic computing approach with MAPE-K, we created a global workspace capable of broadcasting information from various reactive processes into the learning algorithm. The knowledge base within this workspace enables knowledge injection from experts and the generation of new contextual knowledge using semantic web technologies.

Our experimental results demonstrate that the Augmentor framework with context-aware adaptive learning significantly outperforms traditional reactive DAGger, with the performance improvement equal to the knowledge bias introduced by expert contributions to the system.

Beyond the quantitative improvements, our study offers several actionable insights for the design of human-centered AI systems. Drawing inspiration from cognitive theory, we demonstrate how structuring learning architectures around concepts like global accessibility and modular process coordination enables more flexible and scalable adaptation. Context-aware learning further enhances adaptability by softening the rigidity of traditional online algorithms, allowing them to filter, prioritize, and integrate information based on operational relevance and expert input. Most importantly, by including humans in the learning loop—not as passive data providers but as active collaborators—the system fosters a cooperative model of continuous co-evolution, where both humans and machines improve together. These findings suggest that AI systems grounded

in cognitive principles and designed for contextual reasoning and human collaboration are better suited to meet the challenges of dynamic, real-world environments.

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**Data availability** The data that support the findings of this study are proprietary and were provided by Mekanisk Service Halden under a confidentiality agreement. Due to the sensitive nature of the data, which may reveal confidential information about the design of the company's machine parts, the data are not publicly available.

## Declarations

**Conflict of interest** The authors have no relevant financial or non-financial interests to disclose.

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